Transform Methods

1. Periodic f^n's & Fourier series

1.1 Complex valued f^n's

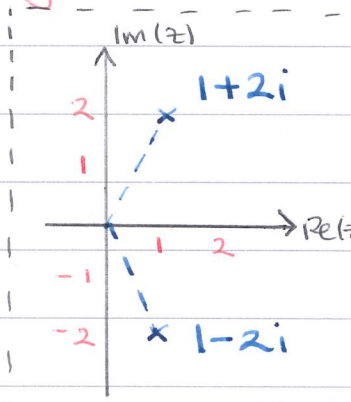
Useful to work w/ complex numbers:

$$z = x + iy \in \mathbb{C}$$

x: Real part $\mathrm{Re}(z)$ — y: Imaginary part $\mathrm{Im}(z)$.

For any $z \in \mathbb{C}$, define:

- $\overline{z} = x - iy$ complex conjugate
- $|z| = \sqrt{x^2+y^2}$ modulus



$|1+2i| = \sqrt{5}$

$$\hookrightarrow \mathrm{Re}(z) = \frac{z+\overline{z}}{2}, \quad \mathrm{Im}(z) = \frac{z-\overline{z}}{2i}$$

$$|z|^2 = z \cdot \overline{z}.$$

Want to consider f^n's $f: \mathbb{R} \to \mathbb{C}$.

Write as

$$f(t) = u(t) + iv(t).$$

Then the derivative is defined as

$$f'(t) = u'(t) + iv'(t).$$

Similarly, the integral of $f(t)$ is

$$\int_a^b f(t)\,dt = F(b) - F(a), \qquad F(t) = U(t) + iV(t)$$

where $\frac{dU}{dt} = u(t)$ & $\frac{dV}{dt} = v(t)$.

Ex 1.1: The complex exponential f^n is defined by

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

This is the usual Maclaurin series for e^z. It can be shown that taking $z = it$ with $t \in \mathbb{R}$

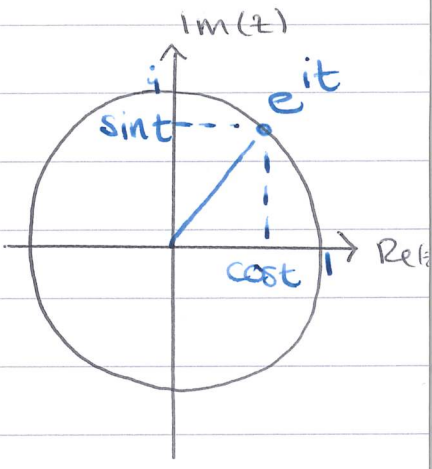
$$e^{it} = \cos t + i \sin t. \quad \text{(Euler's formula)}$$

$\hookrightarrow$ · $|e^{it}| = \sqrt{\cos^2 t + \sin^2 t} = 1$

· $\mathrm{Re}(e^{it}) = \dfrac{e^{it} + e^{-it}}{2} = \cos t$

· $\mathrm{Im}(e^{it}) = \dfrac{e^{it} - e^{-it}}{2} = \sin t$

This tells us that e^{it} is a unit circle inside of $\mathbb{C}$.



Ex 1.2: i.

$$\frac{d}{dt}(e^{itx}) = \frac{d}{dt}(\cos(tx) + i\sin(tx))$$

$$= -x\sin(tx) + ix\cos(tx)$$

$$= xi(\cos t + i\sin t) = xie^{itx}.$$

ii. Take $\alpha = n \in \mathbb{Z}$. Then

$$\int_0^{2\pi} e^{int}\,dt = \left[\frac{e^{int}}{in}\right]_0^{2\pi} = \frac{e^{2\pi in} - e^0}{in} = 0$$

$n \neq 0$

If $n=0$,

$$\int_0^{2\pi} e^{0\cdot it}\,dt = \int_0^{2\pi} dt = 2\pi.$$

$$\rightsquigarrow \int_0^{2\pi} e^{int}\,dt = \begin{cases} 2\pi & n=0, \\ 0 & n\neq 0. \end{cases}$$

1.2 Periodic f^n's & Fourier series

Often deal w/ repeating f^n's:

- Electrical signals
- Sound waves
- Ocean waves.

Def 1.1 A f^n $f:\mathbb{R}\to\mathbb{C}$ is T-periodic if $f(t+T)=f(t)$ for all $t\in\mathbb{R}$.

Ex 1.3:

1. $\sin t$, $\cos t$, e^{it} are 2π-periodic
2. Can rescale to get other periods
   $\rightsquigarrow \sin\left(\frac{2\pi t}{T}\right)$ is T-periodic
3. $$\chi_{\mathbb{Z}}(t) = \begin{cases} 1, & t\in\mathbb{Z} \\ 0, & t\notin\mathbb{Z} \end{cases}$$
   is 1-periodic.

Useful property If f is T periodic then

$$\int_a^{a+T} f(t)\,dt = \int_0^T f(t)\,dt$$

for any $a\in\mathbb{R}$.

↪ Integrals over a period don't depend on startpoint!

For example,

$$\int_{\pi}^{3\pi} e^{it}\,dt = \int_{0}^{2\pi} e^{it}\,dt = 0.$$

Idea: Can we write periodic functions as a sum of 'simpler' functions?

Def 1.2 The (complex) Fourier series of a T-periodic fⁿ f is

$$f(t) \sim \sum_{n=-\infty}^{\infty} c_n e^{in\Omega t}, \qquad \Omega = \frac{2\pi}{T}$$

Read as 'has Fourier series'

where

$$c_n = \frac{1}{T}\int_0^T f(t)\, e^{-in\Omega t}\,dt.$$

↑ Fourier coefficients.

The finite sums

$$S_N(t) = \sum_{n=-N}^{N} c_n e^{in\Omega t}$$

are called the partial sums.

For this defⁿ to make sense, the integrals must be finite. We therefore demand

$$\int_0^T |f(t)|\,dt < \infty$$

$$\rightsquigarrow f \in L^1([0,T]).$$

Right now, this is just a def^{n}, so how does f relate to its Fourier series. For now, assume $f(t)$ can be written as

$$f(t) = \sum_{n\in\mathbb{Z}} c_n e^{in\Omega t}$$

Want to find c_n

$$\Rightarrow e^{-im\Omega t} f(t) = \sum_{n\in\mathbb{Z}} c_n e^{i\Omega t(n-m)}$$

Now integrate over $[0, T]$

$$\int_0^T e^{-im\Omega t} f(t)\,dt = \sum_{n\in\mathbb{Z}} \left(c_n \int_0^T e^{i\Omega t(n-m)}\,dt \right)$$

Change variable to $s = \Omega t$, so

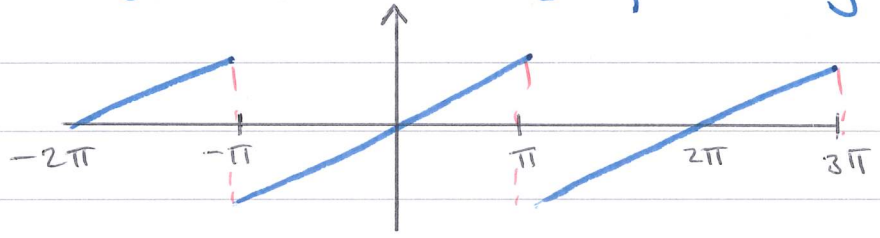
$$\int_0^T e^{-im\Omega t} f(t)\,dt = \frac{1}{\Omega} \sum_{n\in\mathbb{Z}} \left(\int_0^{2\pi} c_n e^{i(n-m)t}\,dt \right)$$

From before, we know that the integral is 0 unless $n = m$. Hence

$$\int_0^T e^{-im\Omega t} f(t)\,dt = \frac{1}{\Omega} \cdot 2\pi c_m = T c_m.$$

$\rightsquigarrow$ Same as def^{n} of Fourier coeff!

Ex 1.4 The sawtooth wave is a 2π-periodic f^{n} defined on $[-\pi, \pi)$ by $f(t) = t$



(6)

$T = 2\pi$, $\Omega = 1$

The Fourier coeffs are

$$c_n = \frac{1}{2\pi}\int_0^{2\pi} f(t)\, e^{-int}\, dt$$

shift int. range

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} t\, e^{-int}\, dt$$

$$= \left[-\frac{te^{-int}}{2\pi in}\right]_{-\pi}^{\pi} + \frac{1}{2\pi in}\int_{-\pi}^{\pi} e^{-int}\, dt \quad (\to 0)$$

$n \neq 0$

$$= -\frac{1}{2\pi in}\left(\pi e^{-in\pi} + \pi e^{in\pi}\right)$$

Using $e^{in\pi} = (-1)^n$, we get

$$c_n = -\frac{1}{2in}\left((-1)^n + (-1)^n\right)$$

$$= \frac{(-1)^{n+1}}{in}, \qquad n \neq 0.$$

For $n = 0$ we have

$$c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} t\, dt = \frac{1}{4\pi}\left[t^2\right]_{-\pi}^{\pi} = 0.$$

Therefore

$$f(t) \sim \sum_{n\neq 0} \frac{(-1)^{n+1}}{in}\, e^{int}$$

Things to think about

- $f(t)$ is real, can check the series is real too
- What happens at $t = \pi$? $f(t)$ is discontinuous there!